

COMPUTER SCIENCE PROJECT (2025-26)

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HARDWARE AND SOFTWARE REQUIRED

➤ **HARDWARE :**

- LAPTOP

➤ **SOFTWARE:**

- IDLE Python 3.12
- MySQL Command line client
- Python-MySQL Connector

HOSPITAL MANAGEMENT SYSTEM

A hospital management system is an information system that manages the aspects of a hospital. This may include the administrative, financial, and medical processing. It is an integrated end-to-end Hospital Management System that provides relevant information across the hospital to support effective decision making for patient care, hospital administration and critical financial accounting, In a seamless flow. This program can look after Inpatients, OPD patients, records, database treatment, status illness, billing etc. it also maintains their hospital info such as ward id, Doctor in charge, Department administering etc.

NEED FOR HOSPITAL MANAGEMENT SYSTEM

1. Minimized documentation and no duplication of records.
2. Reduced paper work.
3. Improved patient care
4. Better Administration Control
5. Faster information flow between various departments
6. Smart Revenue Management
7. Effective billing of various services
8. Exact stock information

INTRODUCTION

The project Hospital Management system includes registration of patients, storing their details into the system, and also computerized billing in the pharmacy, and labs. The software has the facility to give a unique id for every patient and stores the details of every patient and the staff automatically. It includes a search facility to know the current status of each room. User can search availability of a doctor and the details of a patient using the id. The Hospital Management System can be entered using a username and password. It is accessible either by an administrator or receptionist. Only they can add data into the database. The data can be retrieved easily. The interface is very user-friendly. The data are well protected for personal use and makes the data processing very fast. Hospital Management System is powerful, flexible, and easy to use and is designed and developed to deliver real conceivable benefits to hospitals. Hospital Management System is designed for multispeciality hospitals, to cover a wide range of hospital administration and management processes. It is an integrated end-to-end Hospital Management System is a software product suite designed to improve the quality and management of hospital management in the areas of clinical process analysis and activity-based costing.

SYSTEM ANALYSIS

EXISTING SYSTEM :

The current manual system has a lot of paper work. To maintain the records of sale and service manually, is a Time-consuming task. With the increase in database, it will become a massive task to maintain the database. Requires large quantities of file cabinets, which are huge and require quite a bit of space in the office, which can be used for storing records of previous details. The retrieval of records of previously registered patients will be a tedious task. Lack of security for the records, anyone disarrange the records of your system. If someone want to check the details of the available doctors the previous system does not provide any necessary detail of this type. All this work is done manually by the receptionist and other operational staff and lot of papers are needed to be handled and taken care of.

+ ADVANTAGES:

1. No extra training required.
2. Easy to implement.
3. Can be stored anywhere.
4. Requires minimum effort.

+ DISADVANTAGES:

1. Needs lots of paper.
2. Problem with maintenance.
3. Volumes of data becomes problem.
4. Once data is burned it cannot be reproduced easily.
5. Data handling is problem.

PYTHON

SOURCE

CODE

```
import mysql.connector as sql
from tkinter import *

connection=sql.connect(host="localhost",user="root",passwd
="execute",database="hospital_management")
cur=connection.cursor()

a=Tk()

a.geometry("700x700")

a.title("HOSPITAL MANAGEMENT SYSTEM")

a.config(background="#DCFCCA")

label=Label(a,text="WELCOME TO HOSPITAL
MANAGEMENT SYSTEM",font=("Algerian", 20,
"bold"),fg="#0623C4",padx=10,pady=10)

label.pack()

guide1=Label(a,text="CHOOSE AN OPTION")

guide1.pack()


def admin_login():
    passwd=int(input("enter your admin password:"))
    if passwd==1:
        print("login successful")
        print("1. ADD DOCTOR\n2.VIEW DOCTOR\n3. ADD
PATIENT\n4. VIEW APPOINTMENTS")
        no=int(input("Enter the number:"))
```

```

if no==1:
    print("give the following details:")
    id1=int(input("ENTER DOCTOR ID:"))
    name=input("ENTER DOCTOR NAME:")
    spec=input("ENTER SPECIALIZATION:")
    phone=int(input("ENTER PHONE NO.:"))
    email=input("ENTER EMAIL:")
    exp=int(input("ENTER EXPERIENCE (IN YEARS):"))
    ins="INSERT INTO
doctors(ID,Name,Specialization,Phone,Email,Experience)VA
LUES({},'{}','{}',{},'{}',{})".format(id1,name,spec,phone,email,
exp)

    cur.execute(ins)
    print("ADDED")
    connection.commit()

if no==2:
    id1=int(input("enter doctor id:"))
    view1="select * from doctors where id=%s"%(id1,)
    cur.execute(view1)
    data=cur.fetchall()
    print(data)

if no==3:

```

```

print("give the following details:")
id1=int(input("ENTER PATIENT ID:"))
name=input("ENTER PATIENT NAME:")
gender=input("ENTER GENDER ('M' OR 'F'):")
phone=int(input("ENTER PHONE NO.:"))
addr=input("ENTER ADDRESS(CITY,STATE):")
dob=input("ENTER DATE OF BIRTH:")

ins="INSERT INTO
patients(ID,Name,Gender,Phone,Address,DOB)VALUES({},'{{
}}','{{}}',{{}},'{{}}','{{}}')".format(id1,name,gender,phone,addr,dob)

cur.execute(ins)

print("ADDED")

connection.commit()

if no==4:

    app=int(input("enter patient id:"))

    view1="select * from appointments where
id=%s"%(app,)

    cur.execute(view1)

    data=cur.fetchall()

    print(data)

else:

    print("THANK YOU. FOR FURTHER TASKS ,GO TO
MAIN MENU.")

```

else:

print("INCORRECT PASSWORD")

def patient_login():

print("1.for new registration")

print("2.for checking appointments")

print("3. for updating any details:")

p=int(input("enter no. (1/2)"))

if p==1:

print("give the folloing details:")

id_1=int(input("ENTER PAITENT ID:"))

NAME=input("ENTER PAITENT NAME:")

GENDER=input("ENTER GENDER (M/F)")

PHONE=int(input("ENTER PHONE NO. :"))

ADDRESS=input("ENTER YOUR ADDRESS:")

DOB=input("ENTER DATE OF BIRTH :")

ins="INSERT INTO

patients(ID,NAME,GENDER,PHONE,ADDRESS,DOB)VALUE
S({},'{}','{}','{}','{}','{}')".format(id_1,NAME,GENDER,PHONE,ADDRESS,DOB)

cur.execute(ins)

connection.commit()

```
elif p==2:
    ID=int(input("ENTER YOU PATIENT ID:"))
    cur.execute("select * from appointments where
id=%s"%(ID,))
    connection.commit()

elif p==3:
    print("Choose the value you want to
change/n1.ID/n2.NAME/n3.GENDER/n4.PHONE/n5.ADDRE
SS/n6.DOB")

    id1=int(input("enter your patient id:"))
    n=int(input("Enter no. of the value you want to
change:"))

    if n==1:
        new_id=int(input("Enter new id:"))
        query="update patients set ID=%s where ID=%s"
        cur.execute(query, (new_id,id1))
        connection.commit()

    elif n==2:
        name_=input("Enter new name:")
        Q="update patients set NAME=%s where ID=%s"
        cur.execute(Q,(name_,id1))
        connection.commit()

    elif n==3:
```

```
gen=input("Enter correct gender(M/F/other):")
Q2="update patients set GENDER=%s where
ID=%s"
cur.execute(Q2,(gen,id1))
connection.commit()
elif n==4:
    ph=int(input("Enter new phone no. :"))
    Q3="update patients set PHONE=%s where ID=%s"
    cur.execute(Q3,(ph,id1))
    connection.commit()
elif n==5:
    new_add=input("Enter new address:")
    Q4="update patients set ADDRESS=%s here
ID=%s"
    cur.execute(Q4,(new_add,id1))
    connection.commit()
elif n==6:
    new_dob=input("enter correct dob:")
    Q5="update patients set DOB=%s where ID=%s"
    cur.execute(Q5,(new_dob,id1))
    connection.commit()
else:
```

```
print("THANKS FOR VISITING")
```

```
def doctor_login():
```

```
    ID=int(input("Enter doctor id:"))
```

```
    b="select * from appointments where  
doctor_id=%s"%(ID,)
```

```
    cur.execute(b)
```

```
    data=cur.fetchall()
```

```
    for i in data:
```

```
        print(i)
```

```
    print("FOR ANY OTHER UPDATION ,CONSULT  
ADMIN")
```

```
def exit1():
```

```
    print("THANKS FOR VISITING. STAY SAFE STAY  
HEALTHY")
```

```
button = Button(a, text='SIGN UP AS ADMIN')
```

```
button1 = Button(a, text='SIGN UP AS DOCTOR')
```

```
button2 = Button(a, text='SIGN UP AS PATIENT')
```

```
button3 = Button(a, text='EXIT')
```

```
button.pack()
```

```
button.config(command=admin_login)
```

```
button1.config(command=doctor_login)
```

```
button2.config(command=patient_login)
```

```
button3.config(command=exit1)
```

```
button1.pack()
```

```
button2.pack()
```

```
button3.pack()
```

```
a.mainloop()
```

DATABASE

```
mysql> use hospital_management;
```

Database changed

```
mysql> show tables;
```

```
| Tables_in_hospital_management |
```

+

+

| appointments |

doctors

```
| patients
```

```
3 rows in set (0.07 sec)
```

DOCTORS TABLE :

```
mysql> select * from doctors;
```

Id	Name	Specialization	Phone	Email	Experience
101	Ravi	Dentist	9876543211	ravi.dentist@example.com	8
102	Amit	Eye	9876543221	amit.eye@example.com	10
103	Sunil	Gyno	9876543232	sunil.gyno@example.com	12
104	Vijay	ENT	9876543244	vijay.ent@example.com	9
105	Neha	Physician	9876543245	neha.physician@example.com	7
106	Kiran	Orthopedic	9876543526	kiran.ortho@example.com	11
107	Suman	Child	9876543267	suman.child@example.com	5
108	Anil	Skin	9876543287	anil.skin@example.com	6
109	Pooja	Neuro	9876543289	pooja.neuro@example.com	14
110	Raj	Cardio	9876543320	raj.cardio@example.com	15
111	Shreya	Cardio	9898765742	drshreyasingh123@gmail.com	8

APPPOINTMENTS TABLE :

```
mysql> select * from appointments;
```

id	patient_id	doctor_id	appointment_date	reason	status
1	1	101	2025-09-01	Tooth Pain	Completed
2	2	102	2025-09-02	Eye Checkup	Completed
3	3	103	2025-09-03	Pregnancy Care	Pending
4	4	104	2025-09-04	Ear Pain	Completed
5	5	105	2025-09-05	Cough	Completed
6	6	106	2025-09-06	Accident Injury	Pending
7	7	107	2025-09-07	Child Fever	Completed
8	8	108	2025-09-08	Skin Rash	Completed
9	9	109	2025-09-09	Headache	Pending
10	10	110	2025-09-10	Chest Pain	Completed

```
10 rows in set (0.00 sec)
```

PATIENTS TABLE :

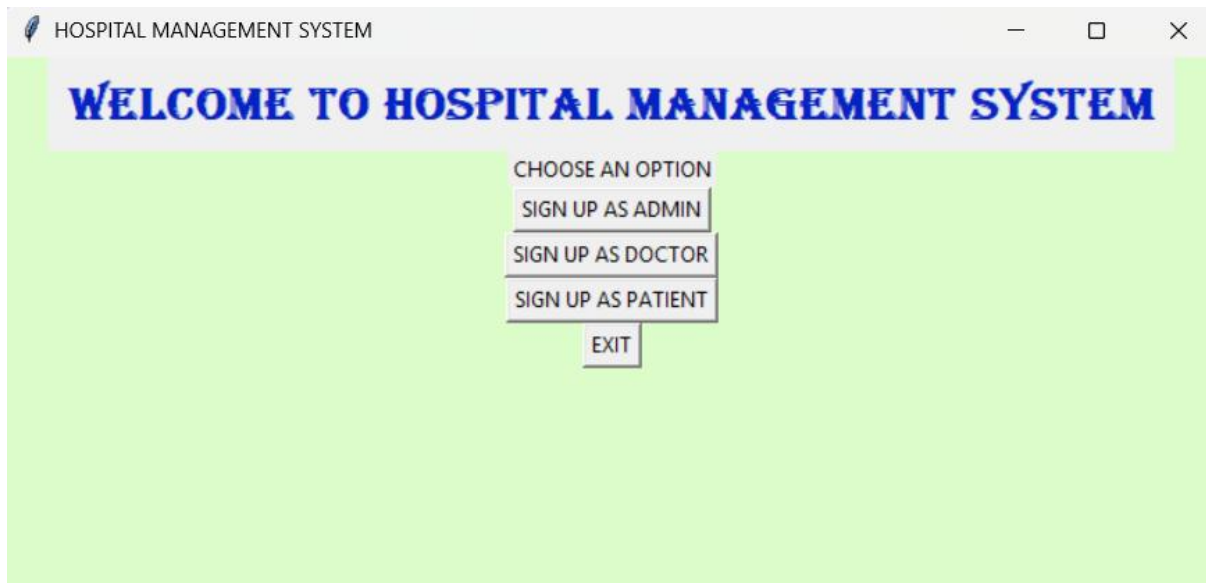
```
mysql> select * from patients;
```

id	name	phone	gender	dob	address
1	Arun	9876500001	Male	1990-05-15	Lucknow, UP
2	Meena	9876500002	Female	1995-08-20	Kanpur, UP
3	Rakesh	9876500003	Male	1988-03-10	Jaipur, Rajasthan
4	Sita	9876500004	Female	2000-07-22	Delhi
5	Vikas	9876500005	Male	1992-11-02	Varanasi, UP
6	Rani	9876500006	Female	1985-09-25	Jodhpur, Rajasthan
7	Manoj	9876500007	Male	1998-01-17	Noida, UP
8	Seema	9876500008	Female	1993-12-12	Udaipur, Rajasthan
9	Ajay	9876500009	Male	1987-04-30	Delhi
10	Kavita	9876500010	Female	1999-06-05	Agra, UP
11	Rahul	9876500011	Male	1991-02-21	Alwar, Rajasthan
12	Priti	9876500012	Female	1996-10-19	Ghaziabad, UP
13	Deepak	9876500013	Male	1994-09-08	Bikaner, Rajasthan
14	Anita	9876500014	Female	2001-11-30	Delhi
15	Karan	9876500015	Male	1997-07-14	Lucknow, UP
16	Shalu	9876500016	Female	1989-08-09	Ajmer, Rajasthan
17	Mohit	9876500017	Male	1990-03-19	Meerut, UP
18	Jyoti	9876500018	Female	1992-12-03	Delhi
19	Suresh	9876500019	Male	1986-05-27	Kota, Rajasthan
20	Geeta	9876500020	Female	1998-09-11	Varanasi, UP

20 rows in set (0.05 sec)

OUTPUTS

1. DISPLAYING MAIN MENU



2. SIGN UP AS ADMIN

```
enter your admin password:1985
login successful
1. ADD DOCTOR
2.VIEW DOCTOR
3. ADD PATIENT
4. VIEW APPOINTMENTS
Enter the number:1
give the following details:
ENTER DOCTOR ID:112
ENTER DOCTOR NAME:RADHIKA
ENTER SPECIALIZATION:SURGEON
ENTER PHONE NO.:9876574823
ENTER EMAIL:drradha234@gmail.com
ENTER EXPERIENCE (IN YEARS):8
ADDED
THANK YOU. FOR FURTHER TASKS ,GO TO MAIN MENU.
```

3. SIGN UP AS DOCTOR

Enter doctor id:105

(5, 5, 105, datetime.date(2025, 9, 5), 'Cough', 'Completed')

FOR ANY OTHER UPDATION ,CONSULT ADMIN

4. SIGN UP AS PATIENT

1.for new registration

2.for checking appointments

3. for updating any details

enter no. (1/2/3): 3

Choose the value you want to change

1.ID

2.NAME

3.GENDER

4.PHONE

5.ADDRESS

6.DOB

enter your patient id:4

Enter no. of the value (numbers are specified above)you want to change:4

Enter new phone no. :9564356788

UPDATED

REFERENCES

In order to work on this project titled-(HOSPITAL MANAGEMENT SYSTEM), the following books and websites are referred by me during the various phases of development of the project.

1) Class 11 th & 12 th Computer Science Books
(SUMITAARORA)

2) www.youtube.com

3) www.python.com

Other than the above-mentioned books, the supervision and suggestions of my teacher and my classmates respectively also helped me to develop this software project.

THANK

YOU